Banking Customer Behavior, Churn and Retention Analytics

Dashboard file: Churn_Dashboard.pbix (Power BI, 5 pages, 15 charts). Analysis notebook: Churn_Analytics.ipynb. Clean data: Churn_Clean_Data.xlsx.

1. Business problem and objectives

The bank has 10,000 customers in this data and 2,038 of them have left. That is a churn rate of 20.4 percent, or one customer in every five. The customers who left were holding 185.7 million in account balances, which is 24.3 percent of all the money the bank held. Management can see that customers leave, but cannot yet see who is leaving, why they leave, or which customers who are still here are likely to leave next.

Core business objectives

- Measure churn and retention for the whole bank and for each customer segment.
- Find the factors that best separate customers who leave from customers who stay.
- Compare the branches on churn and retention.
- Understand how products, activity, balance and tenure relate to loyalty.
- Find high value customers who are still with the bank but show signs of leaving.
- Give management clear actions for targeted retention campaigns.

Data used

Dataset: Bank Customer Churn (Customer-Churn-Records.csv) from Kaggle, uploaded by radheshyamkollipara. It was downloaded without any changes and is kept in the Raw_Data folder. It has one row per customer and 18 columns: credit score, country, gender, age, years with the bank, balance, number of products, credit card, active member, estimated salary, complaint, satisfaction score, card type, reward points and whether the customer left.

What the data does not contain

The assignment asks for branches, time trends and transactions. The data has none of these as separate fields, so the following choices were made and are stated on the dashboard and here:

- Branch: there is no branch column, so each country (France, Germany, Spain) is treated as one branch. This gives 3 branches.
- Time trend: there are no calendar dates. The trend chart uses years with the bank (0 to 10) as the time axis.
- Transactions: there are no transaction rows. Account balance is used for balance behavior and reward points earned is used as a measure of card spending.
- Complaints: 99.5 percent of customers who complained had already left, so a complaint shows an exit that has happened. It is shown in the analysis but is not used in the risk score.

2. Data cleaning, transformation and preparation

All preparation was done in the notebook with simple step by step code. The raw file was in good shape, so the work was mostly checking, renaming and adding useful columns.

Check	Result
Missing values	0
Duplicate rows	0
Duplicate customer ids	0
Age	18 to 92, no odd values
Years with the bank	0 to 10
Credit score	350 to 850
Negative balances	0 (3,617 customers have a zero balance)
Flag columns (card, active, complaint, churn)	Only 0 and 1

```
# Data quality checks on the raw file

Missing values: 0
Duplicate rows: 0
Duplicate customer ids: 0

Age range: 18 to 92
Tenure range: 0 to 10
Credit score range: 350 to 850
Negative balances: 0
Customers with zero balance: 3617
Products owned: [1, 2, 3, 4]
Satisfaction scores: [1, 2, 3, 4, 5]
Countries: ['France', 'Spain', 'Germany']
Card types: ['DIAMOND', 'GOLD', 'SILVER', 'PLATINUM']
Churn flag values: [0, 1]
Complaint flag values: [0, 1]
Active flag values: [0, 1]
Credit card flag values: [0, 1]
```

Output of the data quality checks in the notebook. The raw file has no missing values, no duplicates and every column is inside a sensible range, so no rows had to be removed. All 10,000 customers are kept.

Transformation steps

· Dropped RowNumber (only an index) and Surname (personal information that is not needed).
· Renamed 11 columns to clear names, for example Exited became Churn Flag and Geography became Country.
· Added text labels next to the 0 and 1 columns: Customer Status (Retained or Churned), Activity Status, Credit Card Holder, Complaint Filed.
· Added Branch from the country name.
· Created bands: Age Group, Tenure Group, Balance Band, Credit Score Band, Salary Band, Points Band, and Product Group (single or multiple).
· Created Customer Value: High Value is the top 25 percent of balances (127,644 or more), Low Value is a zero balance, everyone else is Medium Value.
· Created Engagement Segment from activity and satisfaction: Highly Engaged, Engaged, Passive, Disengaged.
· Created a Risk Score and Risk Level for finding customers who may leave (explained below).
· Added hidden order columns so bands sort in the right order in Power BI. The clean data has 41 columns and 10,000 rows.

```
# Customer value and engagement segments

High value cutoff balance: 127644
Customer Value
Medium Value    3883
Low Value       3617
High Value      2500

Engagement Segment
Engaged           3078
Passive           2909
Highly Engaged    2073
Disengaged        1940
```

Counts for the customer value and engagement segments. A quarter of customers (2,500) are High Value, and 1,940 customers are Disengaged (inactive with a low satisfaction score).

Risk score for customers who are still with the bank

Each warning sign that was clearly linked to churn in the analysis adds points. Complaints are left out on purpose, because they follow the exit instead of warning about it.

Warning sign	Points	Churn rate of that group
Age 40 to 59	2	30.8 percent (40 to 49) and 56.0 percent (50 to 59)
Owns 3 or 4 products	2	82.7 percent and 100 percent
Owns only 1 product	1	27.7 percent
Inactive member	1	26.9 percent (active: 14.3 percent)
Germany branch	1	32.4 percent
Female	1	25.1 percent (male: 16.5 percent)
Balance above zero	1	24.1 percent (zero balance: 13.8 percent)

Risk Level: Low is 0 to 2 points, Medium is 3 to 4 points, High is 5 or more points.

```
# Risk level check: share of customers who churned

            count  mean
Risk Level
Low          3830   5.0
Medium       4044  18.1
High         2126  52.4
```

The risk level works. Customers marked Low risk churned 5.0 percent of the time, Medium 18.1 percent and High 52.4 percent. Among customers who are still with the bank, 1,011 are High risk, and 399 of them are also High Value.

3. Interactive dashboard

The dashboard is built in Power BI Desktop and saved as Churn_Dashboard.pbix. It reads Churn_Clean_Data.xlsx. It has 5 pages and 15 charts on pages 1 to 4, plus KPI cards, 5 filters and a drill through page. Every chart answers one business question and has a tooltip with extra numbers.

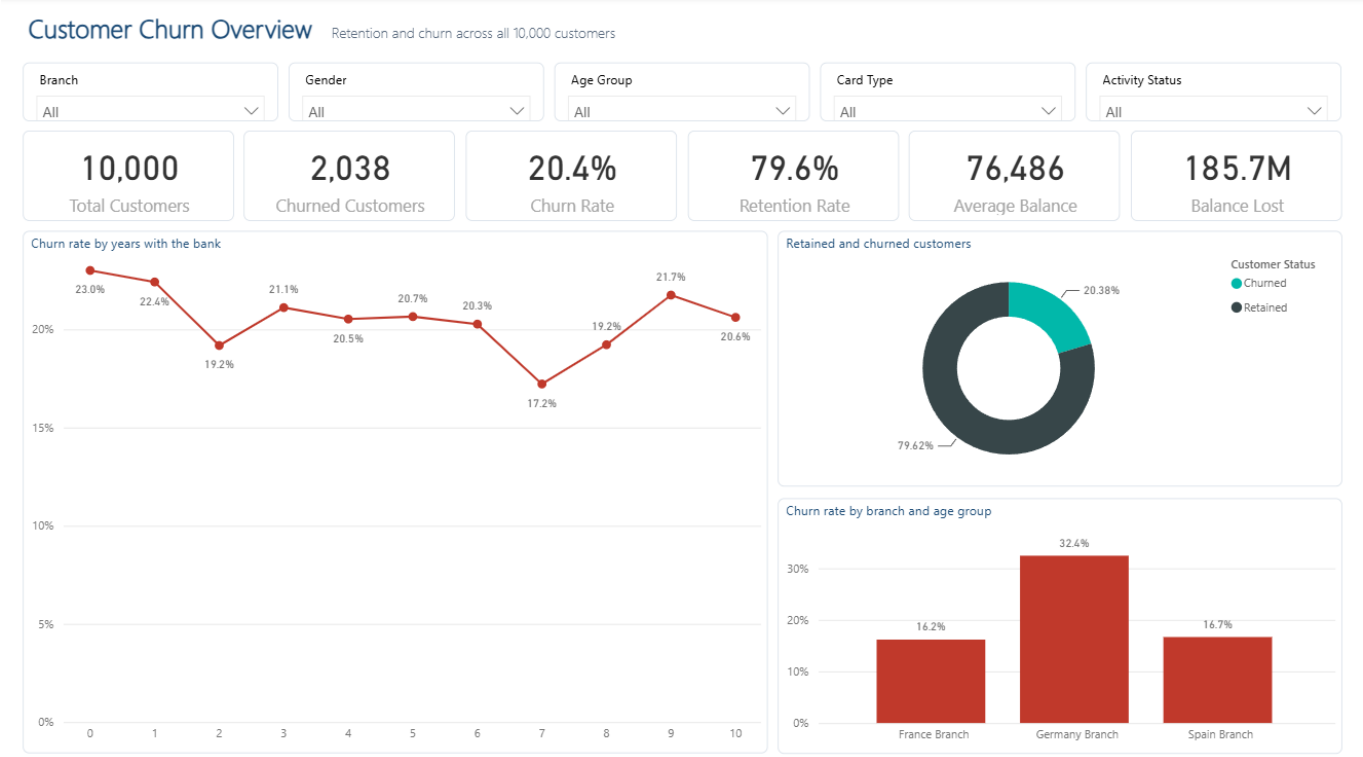
Dashboard features

· KPI cards: total customers, churned customers, churn rate, retention rate, average balance, balance lost, high value customers and high value at risk.
· Filters: Branch, Gender, Age Group, Card Type and Activity Status. They are linked, so one choice applies to every page.
· Drill down: the branch chart drills from Branch to Age Group, the gender chart drills from Gender to Branch.
· Drill through: right click a customer in the at risk table and choose Drill through to open the Customer Detail page.
· Tooltips: every chart shows extra measures such as customers, churned customers, average balance and retention rate.
· Conditional formatting: churn rate is colored green to red in the branch table, and the risk score is shaded in the at risk table.
· Time based trend: churn rate by years with the bank (the data has no calendar dates).

Main measures (DAX)

Measure	Meaning
Churn Rate	Churned Customers divided by Total Customers
Retention Rate	Retained Customers divided by Total Customers
Balance Lost	Total balance held by customers who churned
High Value Churn Rate	Churn rate for customers in the top 25 percent of balances
At Risk Customers	Retained customers with a High risk level
High Value At Risk	Retained customers who are High Value and High risk
Balance At Risk	Total balance held by High Value At Risk customers
Inactive High Value	Retained High Value customers who are inactive members
Churn vs Overall	Churn rate of a segment minus the overall churn rate

Page 1: Overview



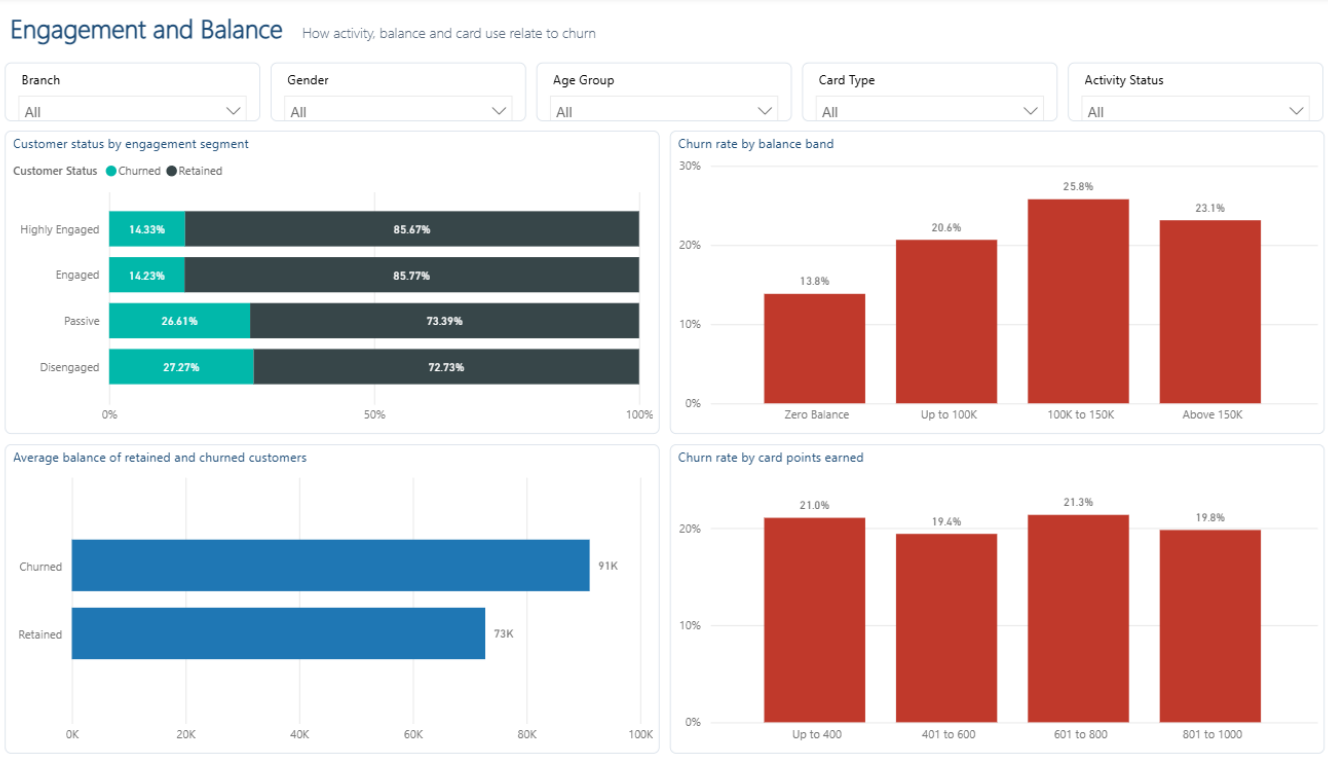
Overall view. The bank keeps 79.6 percent of its customers and loses 20.4 percent. The line chart shows churn by years with the bank: it stays between 17 and 23 percent for every tenure year, so tenure has almost no effect. The branch chart shows Germany at 32.4 percent against 16.2 and 16.7 percent for France and Spain. Answers questions 1, 3 and 6.

Page 2: Segments and Products



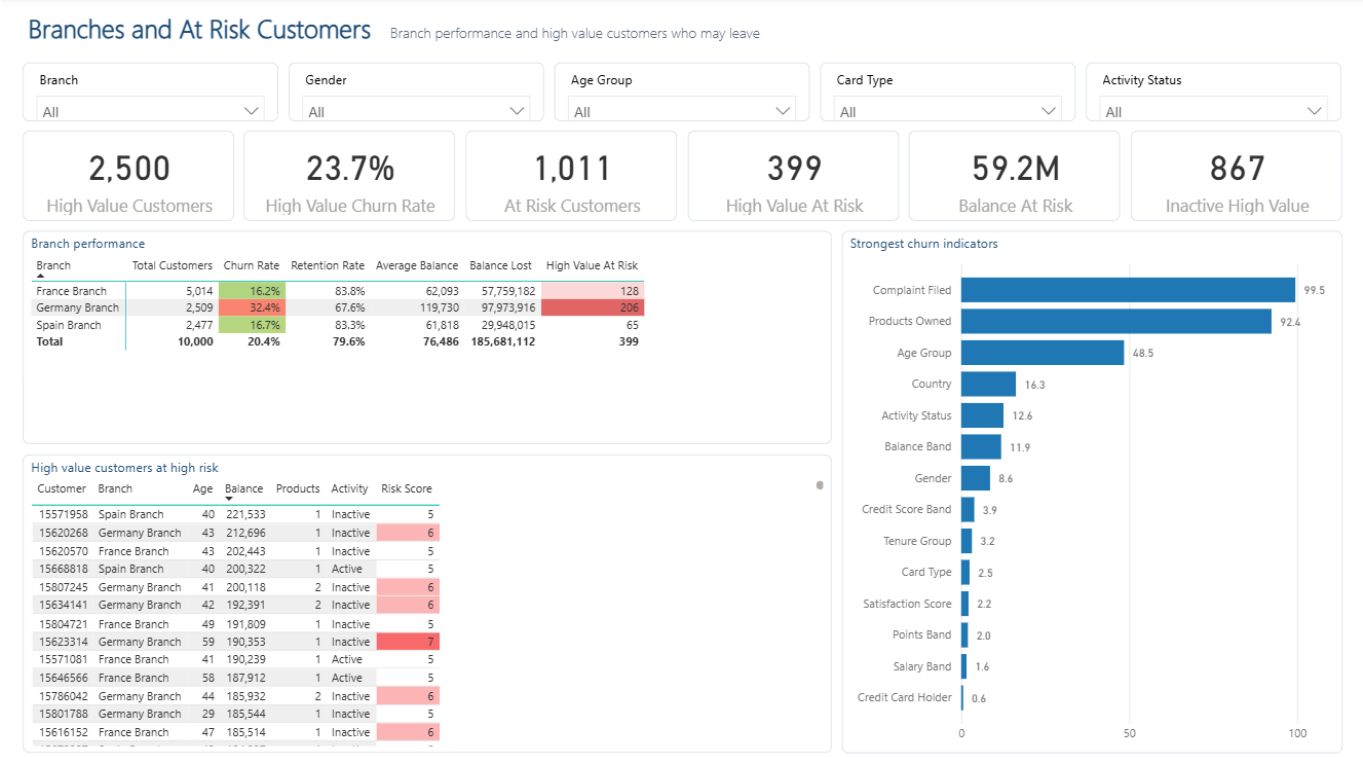
Churn rises sharply with age, from 7.6 percent for customers under 30 to 56.0 percent for ages 50 to 59. Women churn more than men (25.1 against 16.5 percent). Card type makes little difference. Customers with 2 products churn the least (7.6 percent), but 3 and 4 products churn 82.7 and 100 percent. Half of all customers (50.8 percent) own only one product. Answers questions 2, 4, 5 and 9.

Page 3: Engagement and Balance



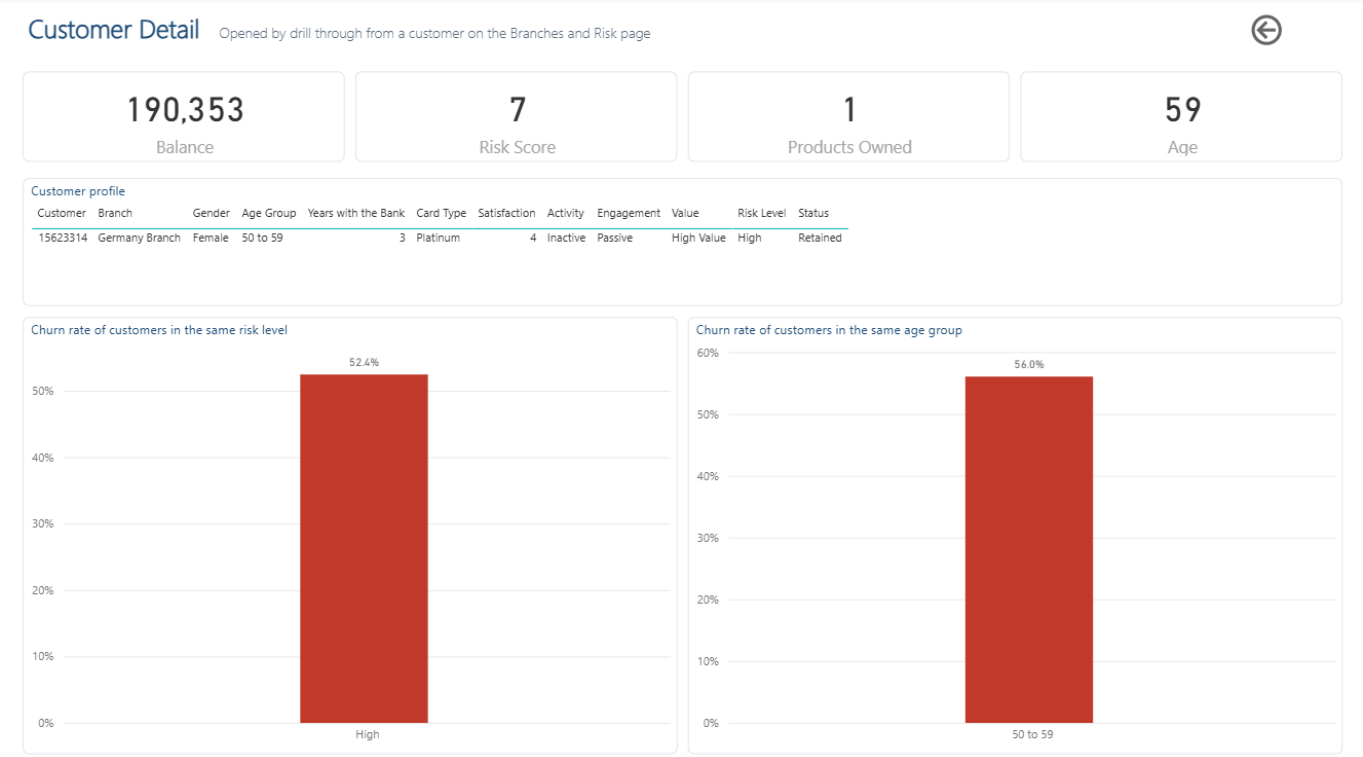
Passive and Disengaged customers churn at about 27 percent, almost double the 14 percent of Engaged customers. Customers with a balance churn more than customers with a zero balance, and the 100K to 150K band is the highest at 25.8 percent. Churned customers had a higher average balance (91K) than retained customers (73K), so the bank loses its richer customers. Card points (spending) show no clear pattern. Answers questions 7, 8 and 11.

Page 4: Branches and At Risk Customers



The Germany branch holds 2,509 customers, has the highest churn (32.4 percent) and the highest average balance, so it lost 98.0 million, more than half of all the balance lost. 399 High Value customers are still here but have a High risk level, holding 59.2 million. The bar chart ranks the factors by the gap in churn rate between their best and worst group. Answers questions 6, 10, 12 and 13.

Page 5: Customer Detail (drill through)



Opened by right clicking a customer on page 4. It shows one customer's balance, risk score, products, age and profile, and compares with the churn rate of all customers in the same risk level and age group. A relationship manager can use it to prepare a call.

4. Analysis by topic

Customer demographics and segmentation

Age is the strongest segment factor after products. Churn is 7.6 percent under 30, 10.9 percent for 30 to 39, 30.8 percent for 40 to 49, 56.0 percent for 50 to 59 and 27.9 percent for 60 and above. Women churn at 25.1 percent against 16.5 percent for men. Salary band and credit score band hardly matter (all between 19.7 and 23.6 percent).

Churn and retention

```
# Q1 overall churn and retention

Total customers: 10000
Churned customers: 2038
Retained customers: 7962
Churn rate: 20.4
Retention rate: 79.6
```

Overall churn is 20.4 percent and retention is 79.6 percent.

Product ownership and adoption

```
# Q4 and Q5 churn by number of products

                Customers  Churned  Churn Rate  Retention Rate
Products Owned
1                    5084     1409        27.7            72.3
2                    4590      349         7.6            92.4
3                     266      220        82.7            17.3
4                      60       60       100.0             0.0
```

Customers with 2 products are the most loyal. Having more than 2 products is not a sign of loyalty here: 266 customers own 3 products and 82.7 percent left, and all 60 customers with 4 products left. Single product customers (50.8 percent of the bank) churn at 27.7 percent, so the second product is the opportunity. Credit card ownership (20.2 against 20.8 percent) and card type make almost no difference.

Customer activity and engagement

Inactive members churn at 26.9 percent against 14.3 percent for active members, and 48.5 percent of customers are inactive. Satisfaction score does not separate leavers from stayers (19.6 to 21.8 percent for every score), so the survey score alone is not an early warning.

Account balance and transaction behavior

Zero balance customers churn least (13.8 percent). Customers with balances of 100K to 150K churn most (25.8 percent). The average balance of churned customers is 91,109 against 72,743 for retained customers. Reward points, used here as a measure of card spending, show no relation to churn (19.4 to 21.3 percent in every band).

Customer tenure and churn

Tenure has almost no relationship with churn. New customers (0 to 1 years) churn at 22.6 percent and loyal customers (8 to 10 years) at 20.5 percent. Churn is a problem at every stage, not only in the first year.

Branch wise churn and retention

Germany Branch: 32.4 percent churn, 67.6 percent retention, average balance 119,730. Spain Branch: 16.7 percent churn, 83.3 percent retention. France Branch: 16.2 percent churn, 83.8 percent retention. Germany has a quarter of the customers but 40 percent of the leavers (814 of 2,038).

High value and at risk customers

```
# High value customers at high risk (top rows)

High value customers at risk: 399
Balance held by them: 59211157
```

	Customer ID	Branch	Age	Balance	Products Owned	Activity Status	Risk Score	Risk Level
0	15623314	Germany Branch	59	190353.08	1	Inactive	7	High
1	15724336	Germany Branch	49	171601.27	1	Inactive	7	High
2	15590327	Germany Branch	42	166031.45	1	Inactive	7	High
3	15764432	Germany Branch	42	164307.77	1	Inactive	7	High
4	15569764	Germany Branch	41	154007.21	1	Inactive	7	High
5	15714087	Germany Branch	45	151855.33	1	Inactive	7	High
6	15655574	Germany Branch	40	150777.10	1	Inactive	7	High
7	15639552	Germany Branch	40	148897.02	1	Inactive	7	High
8	15778320	Germany Branch	40	148495.64	1	Inactive	7	High
9	15806049	Germany Branch	49	140510.89	1	Inactive	7	High

The 10 highest balance customers who are still with the bank and have a High risk level. Most are in the Germany Branch, own one product and are inactive. In total 399 High Value customers are at risk (59.2 million), and 867 High Value customers are inactive members (128.8 million).

5. Answers to the business questions

Question	Answer
1. Overall churn and retention?	Churn 20.4 percent, retention 79.6 percent (2,038 of 10,000 left).
2. Which segments churn most?	Ages 50 to 59 (56.0 percent), Germany (32.4 percent), ages 40 to 49 (30.8 percent), customers with 3 or 4 products, inactive members (26.9 percent).
3. Does tenure matter?	No. Churn is 17 to 23 percent in every tenure year.
4. Which products go with higher retention?	Holding 2 products (92.4 percent retained). Card type and credit card ownership make little difference.
5. Do multiple products mean lower churn?	Only for 2 products. With 3 or 4 products churn jumps to 82.7 and 100 percent.
6. Which branches churn most?	Germany Branch (32.4 percent). France 16.2 and Spain 16.7 percent.
7. Activity of retained and churned customers?	55 percent of retained customers are active against 36 percent of churned. Satisfaction and points are the same for both groups.
8. Balance and churn?	Customers with a balance churn more (24.1 percent) than zero balance customers (13.8 percent). Churned customers had a higher average balance.
9. Which demographics churn more?	Older customers (40 to 59), women, and customers in Germany. Salary and credit score hardly matter.
10. High value but disengaged?	867 High Value customers are inactive members holding 128.8 million. 399 are High Value and High risk, holding 59.2 million.
11. Patterns among churned customers?	Older (average age 44.8 against 37.4), more often in Germany (39.9 percent of leavers against 21.3 percent of stayers), more often women (55.9 percent), mostly inactive (64 percent).
12. Who to prioritize?	Germany ages 40 to 59 (528 leavers), France ages 40 to 59 (515 leavers), customers with 1 product who are inactive, and the 399 High Value customers at risk.
13. Strongest indicators?	Number of products, age and country, then activity and balance. Complaints are almost always filed by customers who already left.
14. How to improve retention?	See the recommendations in section 6.

```
# Q13 strongest churn indicators

                Factor       Highest Group  High Rate               Lowest Group  Low Rate   Gap
0      Complaint Filed                 Yes       99.5                         No       0.1  99.5
1       Products Owned                   4      100.0                          2       7.6  92.4
2            Age Group           50 to 59       56.0                  Under 30       7.6  48.5
3              Country             Germany       32.4                     France      16.2  16.3
4      Activity Status            Inactive       26.9                     Active      14.3  12.6
5         Balance Band         100K to 150K       25.8               Zero Balance      13.8  11.9
6               Gender               Female       25.1                       Male      16.5   8.6
7    Credit Score Band          500 or less       23.6                 Above 800      19.7   3.9
8         Tenure Group  New (0 to 1 years)       22.6  Established (5 to 7 years)      19.4   3.2
9            Card Type              Diamond       21.8                       Gold      19.3   2.5
10  Satisfaction Score                    2       21.8                          3      19.6   2.2
11         Points Band          601 to 800       21.3                 401 to 600      19.4   2.0
12         Salary Band         150K to 200K       21.5                 50K to 100K      19.9   1.6
13  Credit Card Holder                   No       20.8                        Yes      20.2   0.6
```

Gap in churn rate between the highest and lowest group of each factor. Products, age and country have the biggest gaps once complaints are set aside.

6. Executive summary and recommendations

Summary

The bank loses 1 customer in 5 and the lost customers held 185.7 million. Churn is not spread evenly. It is concentrated in customers aged 40 to 59, in the Germany branch, in customers with 1 product or with 3 to 4 products, and in inactive members. Tenure, salary, credit score, card type, satisfaction score and card spending do not explain churn. The bank also loses richer customers: churned customers held on average 25 percent more than retained customers. Today 1,011 customers who are still here have a High risk level, and 399 of them are High Value, holding 59.2 million.

Recommendations

· Start a retention programme for customers aged 40 to 59, beginning with the Germany branch. Germany ages 40 to 49 and 50 to 59 alone lost 528 customers, with churn of 45.1 and 70.0 percent.

· Review the Germany branch. Its churn is twice that of France and Spain. If it matched the other two branches (about 16.5 percent) the bank would keep about 400 more customers. Check service quality, fees, local competition and staffing.

· Call the 399 High Value customers who are at risk (59.2 million). Use the At Risk table on page 4 and the Customer Detail page to prepare each call. Offer a relationship manager, a fee review or a better rate.

· Reactivate the 867 inactive High Value customers (128.8 million) with personal outreach and offers that give a reason to use the account.

· Cross sell a second product to single product customers. Customers with 2 products churn 7.6 percent against 27.7 percent for 1 product, and half of the bank has only 1 product.

· Investigate the 3 and 4 product bundles. 326 customers hold them and 86 percent of them left. Check pricing, complexity and mis selling before pushing these bundles.

· Fix complaint handling. 99.5 percent of customers who filed a complaint left. Resolve complaints within days, escalate the ones from High Value customers, and record the complaint date so it can warn the bank early.

· Look at the service experience for women, who churn at 25.1 percent against 16.5 percent for men.

· Do not spend campaign money on card type upgrades or points offers. They show no link to churn.

· Improve the data: add account open dates, transaction records and branch codes so that the bank can track churn by month and build an early warning model.

Limits of this analysis

The data is a single snapshot without calendar dates, so trends over calendar time cannot be shown. The 3 branches are countries. The risk score is a simple rule based score built from the same data, so it should be checked against new customers before it is used for large decisions.